

Graphs

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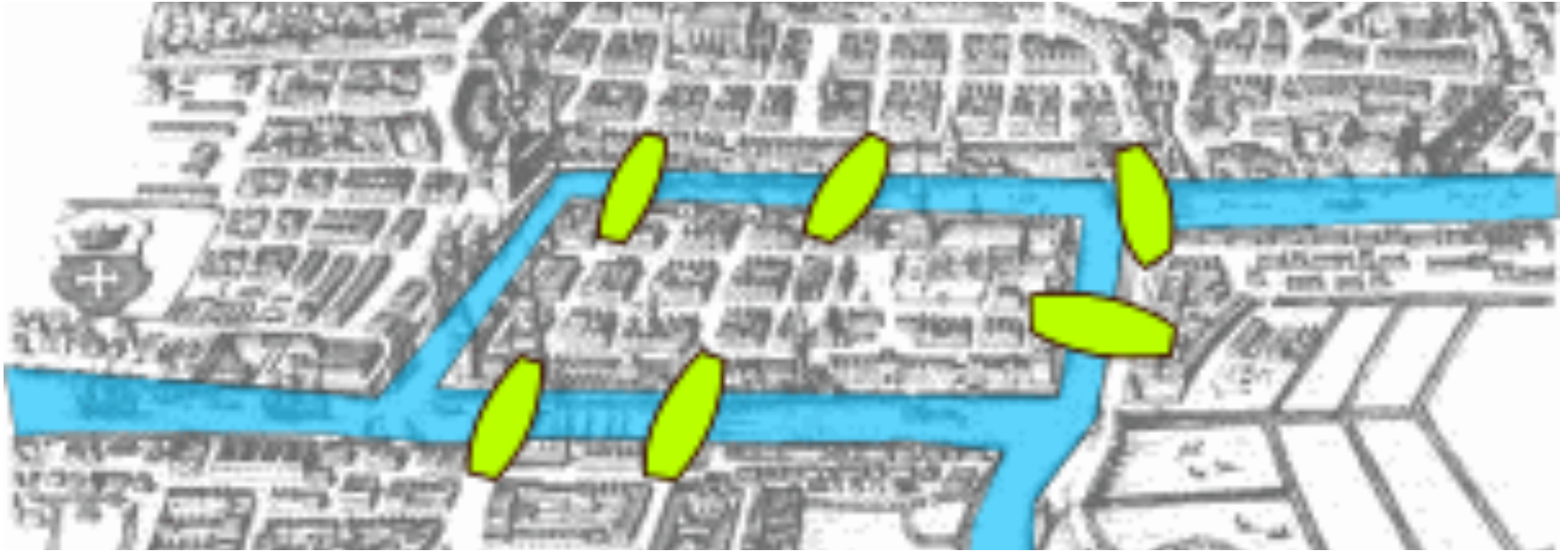
Graph Theory

- Father of graph theory: Leonhard Euler
- Swiss mathematician
- *Seven Bridges of Königsberg, 1736*



Leonhard Euler (source: Wikipedia)

Seven Bridges of Königsberg

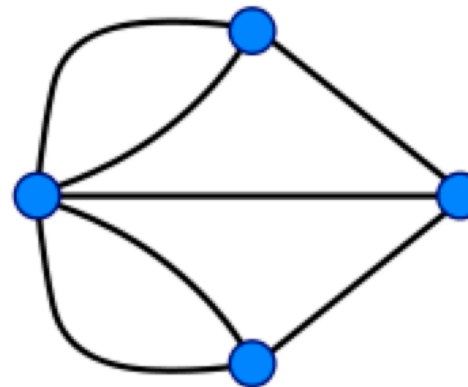
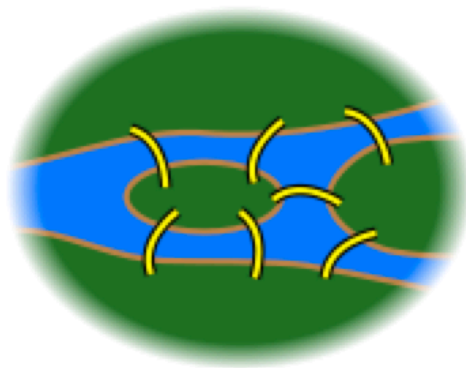
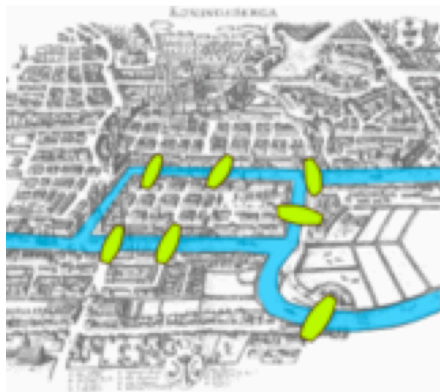


Is there a walk that traverses each bridge exactly once?

What is a graph?

- Vertices and edges.
- Nodes and links.
- People and relationships.

Seven Bridges of Königsberg



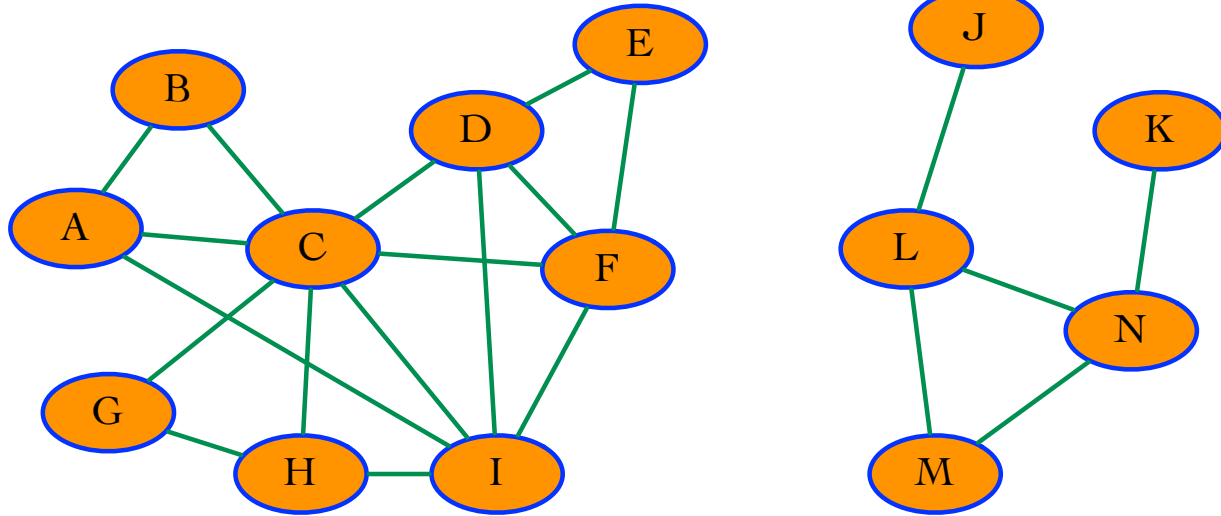
Theorem

There is a walk through a graph that traverses each edge exactly once if and only if the graph is connected and there are exactly two or zero vertices of odd degree.

Types of Networks

- Collaboration networks
- Who-talks-to-whom graphs
- Information linkage graphs
- Technological networks
- Biological networks

Graph Basics



Definition

A vertex A and a vertex B are neighbors if there is an edge AB between A and B.

D is neighbors with E, F, and C, but not B.

Basic Graph Representations

There are two basic ways to represent a graph, $G = (V, E)$, $V = \{v_1, v_2, \dots, v_n\}$:

1. An *adjacency matrix* is an $n \times n$ array where the (i, j) entry is:

$a_{ij} = 1$ if there is an edge from v_i to v_j .

$a_{ij} = 0$ otherwise.

2. An *adjacency list* is a set of n linked lists, one for each vertex.

The linked list for vertex v holds the names of all vertices u such that there is an edge from v to u .

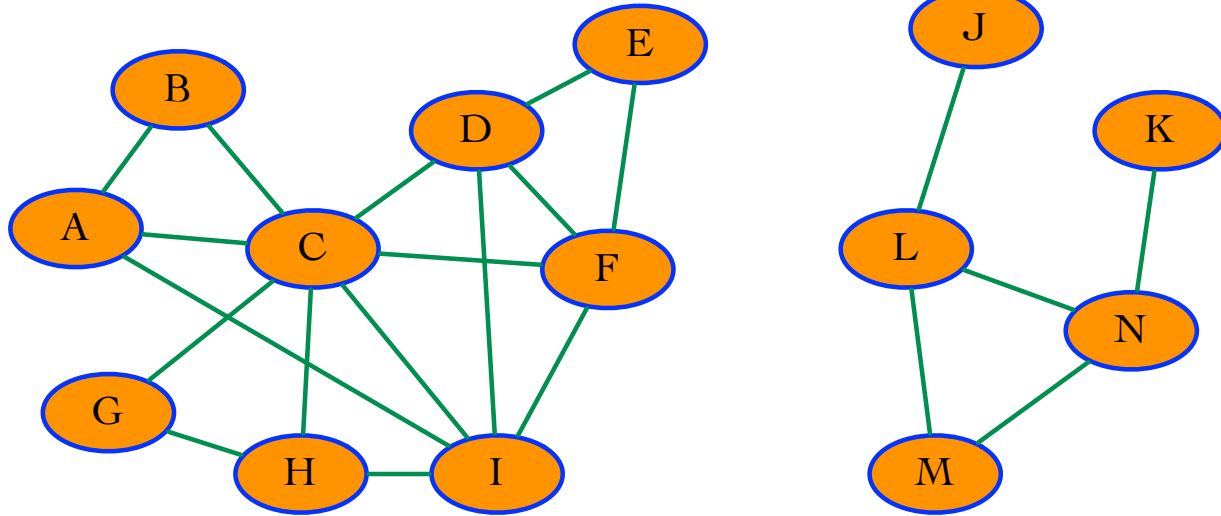
- What is the size of each data structure?
- How long does it take to find a particular edge for each data structure?
- How long does it take to iterate over all edges adjacent to a given vertex?

Graph Representation Tradeoffs

The best data structure depends on your data, the type of queries you will run and other factors

- On hash tables: despite “better” asymptotics, it is not always the best solution:
 - Not very cache friendly
 - Tends to have larger constant factors for both time and space
 - Operation times may vary
 - May introduce randomness in the order of visits
- For this class: your pseudocode should be written at a higher level of abstraction, independent of representation. For example, if you want to iterate over the neighbors of a vertex:
 - **Don't write:**
 - **While** pointer not null: pointer $\leftarrow$ pointer.next (adjacency list)
 - **For** j in range(n): neighbor $\leftarrow$ matrix[i,j] (matrix)
 - **Do write:**
 - **For all** u neighbor of v:
- In the real world: these factors may absolutely matter, but analysis out of scope for this class
- You may also want to use different representations for storage and for different applications

Basic Ways to Describe a Graph

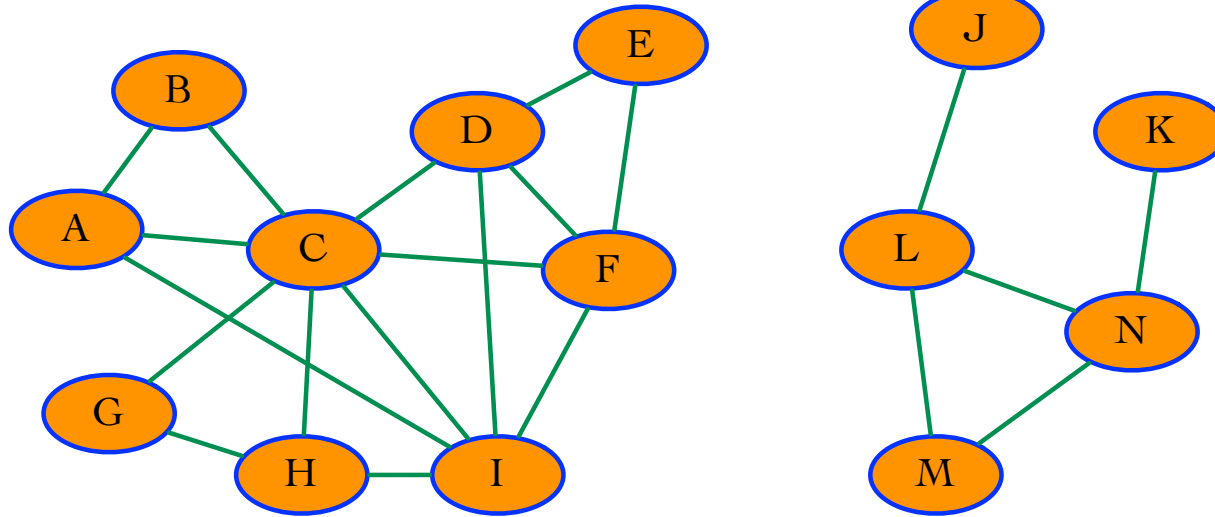


Definition

The degree of a vertex is the number of edges adjacent to it (or the number of neighbors).

C has degree 7. J has degree 1.

Graph Basics



Definition

The degree distribution of a graph is the number of vertices of each degree.

$\{0, 2, 4, 4, 2, 1, 0, 1\}$

or $\{0, 1/7, 2/7, 2/7, 1/7, 1/14, 0, 1/14\}$ (normalized)